



MTH101, Basic Linear Algebra

Monsoon 2025

11/10/2024, 5:30 PM-7:00 PM.

Midterm Examination

- (1) (5pts) Find the inverse of the matrix

$$S = \begin{pmatrix} 1 & 2 & -1 \\ -2 & 0 & 1 \\ 1 & -1 & 0 \end{pmatrix}$$

using the reduced row echelon method.

- (2) (5pts) Prove or disprove the following statements

(i) For any two $n \times n$ matrices A and B , $A^2 - B^2 = (A + B)(A - B)$

(ii) If A is an $n \times n$ matrix such that $A^2 = 0$ then $A + I_n$ is invertible.

- (3) (5pts) Find the equation of the line which passes through the point $(1, 2)$ but does not intersect the line $X - Y = 0$. Justify your answer.

- (4) (5pts) Let x be a nonzero number. Verify using mathematical induction that

$$\frac{x^{n+1} - 1}{x - 1} = 1 + x + x^2 + \cdots + x^n, \quad \text{for all positive integers } n.$$

- (5) (5pts) Find the set of all solutions of the following system of equations

$$X_1 + 6X_2 + 4X_3 + 2X_4 = 0$$

$$X_1 + X_2 + X_3 - X_4 = 1$$

$$X_2 + X_3 + X_4 = 0$$

by reducing the corresponding augmented matrix to its reduced row echelon form.